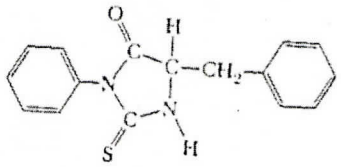
Cume Exam: Biochemistry March 2, 2018

1) Amino acid analysis of the an oligopeptide, consisting of 7 residues, gave the following data:

Asp Leu Lys Met Phe Tyr

The following biochemical data were observed:

a. Trypsin treatment had no apparent effect.
b. The phenylthiohydantoin released by Edman degradation was



c. Brief chymotrypsin treatment yielded several products, including a dipeptide and a tetrapeptide. The amino acid compositions of the tetrapeptide was Leu, Lys, and Met.
d. Cyanogen Bromide treatment yielded a dipeptide, a tetrapeptide and free Lys.

What is the sequence of this heptapeptide? Please write out the sequence. Also, please comment on each of the findings of the biochemical data in order from a-d. (40 points)

2. Which form of chromatography would be the most efficient method to separate the following groups of peptides? (30 points)

a) Peptide 1- LKRYGRRHNPK
   Peptide 2- LKNFGNSCFPK

b) Peptide 1- AGYLKEDNCSTLI
   Peptide 2- CAYLG

c) Peptide 1- (His_6)-YTMNPGYTR
   Peptide 2- RTNPGNYTR

3. Given the following data, deduce the original amino acid sequence from which they were derived (30 points):

a) Aminoethylation followed by trypsin:
- (Ala, Arg, Gly, Met, Phe)
- 2(AECys)
- (Ala, AECys, Phe)
- (Asp, Lys, Tyr)

b) CNBr treatment:
- (Ala2, Asp, Arg, Cys3, Gly, Lys, Met, Phe2, Tyr)

c) Chymotrypsin treatment:
- (Ala, Gly, Lys, Met, Phe)
- (Asp, Cys, Tyr)
- (Ala, Arg, Cys_2, Phe)

d) CNBr plus Trypsin:
- (Gly, Met)
- (Ala, Asp, Cys_3, Lys, Phe, Tyr)
- (Ala, Arg, Phe)